William Mallard

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Systems engineer and computational biologist with 20 years of experience building high-performance computational systems for scientific research — from FPGA signal processing to large-scale genomics to protein biophysics. Seeking roles at the intersection of computational engineering and biology: ML infrastructure, GPU-accelerated scientific computing, and computational biology.

Education

Harvard University — *Cambridge, MA*
PhD, Biochemistry — *8/2016 – 5/2025*

University of California, Berkeley — *Berkeley, CA*
BSc, Engineering Physics; *minors in Computer Science and Mathematics* — *1/2002 – 5/2007*

Skills

Languages: Python, C, C++, CUDA, Assembly, Java, Matlab, Simulink, Bash

Frameworks: PyTorch, NumPy, SciPy, Pandas, scikit-learn, scikit-image

Systems: Linux, Xilinx FPGAs, GPUs, HPC/SLURM

Domains: Machine Learning, Computer Vision, Structural Biology, Genomics, Scientific Computing

Projects

bacterial phospho-site predictor (2026) — Single-layer MLP on ESM-2 embeddings; matches the performance of a published ten-feature ensemble; identified train-test leakage inflating their benchmark.

gpu-xengine (2026) — Ported my FPGA-based ISI correlator X-engine (from Wishnow et al. 2010) to CUDA, increasing its throughput 1.5× by restructuring data movement (memory coalescing, memory pinning).

twitter-news MCP server (2026) — 18-tool RAG interface for LLM-driven fuzzy/semantic search, clustering, and discourse analysis over an archive of ∼1M tweets. Ingestion pipeline: Twitter API to PostgreSQL, with image OCR (via Tesseract), local translation (via NLLB-200), and embedding (via Qwen3 on MLX).

Experience

Harvard University — *Cambridge, MA*
Graduate Student Researcher, Molecular and Cellular Biology — *8/2016 – 12/2025*

Led a multi-year research project from conception through publication, discovering a novel regulatory mechanism in bacterial cell division (PLoS One, 2025).

- **Structural modeling of IDRs:** Sampled 5,000 AlphaFold Multimer conformations of an IDR binding a globular domain; built a contact map clustering pipeline that identified candidate binding modes and contact residues; validated these predictions experimentally via NMR and ITC
- **Structural analysis:** Built pipelines for phase optimization, peak normalization, and comparative intensity analysis of 2D NMR spectra (NOESY, HSQC); walked spectra across a phospho-mutant allelic series to identify the structural mechanism by which phosphorylation disrupts binding
- **Computer vision pipelines:**
  - Built end-to-end image analysis pipelines; hand-wrote core algorithms in vectorized NumPy; profiled to eliminate bottlenecks; parallelized processing on a SLURM cluster
  - *Bacterial morphometry:* cell segmentation, active contour refinement, Voronoi skeletonization
  - *Z-ring constriction dynamics:* flat-field correction, frame registration, kymograph analysis
  - *Z-ring treadmilling velocity:* particle tracking, MSD curve fitting
- **Experimental biology:**
  - *Live-cell microscopy:* fluorescence, TIRF, and confocal imaging; custom filter cube design
  - *Genetic engineering:* designed and built multi-fragment markerless allelic replacements

- **Scientific writing:** Independently managed the entire writing, revision, and peer review process as corresponding author (6/2025–12/2025).

Tools: Python, NumPy, Pandas, SciPy, scikit-image, Fiji, Jython, AlphaFold, SLURM, Bash, NMRPipe, Poky

Broad Institute of MIT and Harvard *Cambridge, MA*
Computational Biologist, Rinn Lab *2/2013 – 7/2016*

Contributed genomics analyses to 13 publications in stem cell biology and developmental epigenetics.

- Built end-to-end data processing pipelines, handling ∼10TB of data across 20+ projects
- Developed custom toolchains for multi-omics integration: RNA-seq, ChIP-seq, ATAC-seq, bisulfite sequencing; differential expression, differential binding, peak-to-gene mapping
- Wrote a Python C extension for automated job monitoring and queue management on LSF clusters

Tools: Python, NumPy, Pandas, SciPy, scikit-learn, C, LSF, Unix pipelines

Broad Institute of MIT and Harvard *Cambridge, MA*
Software Engineer, The Cancer Genome Atlas *4/2012 – 2/2013*

Maintained and extended the GDAC Firehose pipeline, a production genomics platform processing tens of thousands of samples across dozens of cancer types.

- Triaged and resolved pipeline failures across automated analysis runs, diagnosing performance bottlenecks in data ingestion and processing stages
- Rewrote memory-intensive R scripts into streaming Python, enabling prototype pipelines to scale to tens of thousands of samples
- Identified a Python standard library limitation blocking large-dataset processing; contributed a fix to CPython 3.4 in zipfile and shutil (issues #17189, #17201)

Tools: Python, NumPy, R, Bash

Space Sciences Laboratory *Berkeley, CA*
Systems Engineer, Townes Lab *9/2009 – 8/2011*

Built a real-time FX correlator for the Infrared Spatial Interferometer at Mount Wilson Observatory.

- Designed a 138 Gbps signal processing system across 3 synchronized Xilinx Virtex-5 FPGAs, channelizing 2.88 GHz of bandwidth into 64 spectral channels with real-time cross-correlation
- Achieved a 1.8× clock speedup (200 → 360 MHz) over existing CASPER designs by mapping core FFT arithmetic onto DSP48Es and developing a BRAM/DSP floorplanning methodology
- Contributed 50+ optimized DSP commits upstream to CASPER's open-source FPGA library; documented floorplanning methodology in CASPER Memo #42
- Built an end-to-end data acquisition pipeline, from FPGA fabric to PowerPC firmware to Python control software and networked storage
- Instrument resolved individual molecular signatures in Betelgeuse's atmosphere; results published in Wishnow et al. 2010 (SPIE); code and documentation available on GitHub.

Tools: Simulink, MATLAB, C, Python

Center for Astronomy Signal Processing and Electronics Research *Berkeley, CA*
Systems Engineer, Werthimer Lab *4/2008 – 9/2009*

Contributed to CASPER, an open-source FPGA toolkit for radio astronomy signal processing.

- Developed a 10 Gbps UDP receiver with a synchronized ring buffer; core architecture adopted into PySPEAD, the data transport layer for the Square Kilometre Array
- Refactored and extended Simulink blocks and mask scripts in the CASPER FPGA library
- Built enclosure and RF signal routing for a spectrometer deployed at Arecibo Observatory

Tools: Simulink, MATLAB, C, Python

MusicianLink *San Jose, CA*
Software Engineer *6/2007 – 3/2008*

Designed the binary protocol and connection manager for a real-time networked music device.

UC Berkeley Clustered Computing *Berkeley, CA*
Unix Systems Administrator *1/2006 – 8/2006*

Administered Unix systems, server hardware, and job schedulers for scientific HPC clusters.

Massachusetts Institute of Technology *Cambridge, MA*
Research Assistant, Chuang Lab *6/2005 – 8/2005*

Wrote pulse sequencer firmware in microcontroller assembly to characterize quantum ion traps.

University of California, Berkeley *Berkeley, CA*
Research Assistant, Holzapfel Lab *6/2004 – 5/2005*

Wrote an instrument control system in C++ to thermocycle a cryogenic helium dilution fridge.

Selected Publications

Mallard WJ, Pham VV. "FtsZ phosphorylation modulates tail-core binding to tune cell division in *Bacillus subtilis*." ***PLoS One*** (2025).

Choudhuri A, Trompouki E, Abraham BJ, Colli LM, Kock KH, **Mallard W**, . . . , Rinn J, Albert PS, Bulyk ML, Chanock SJ, Young RA, Zon LI. "Common variants in signaling transcription-factor-binding sites drive phenotypic variability in red blood cell traits." ***Nature Genetics*** (2020).

Kaewsapsak P, Shechner DM, **Mallard W**, Rinn JL, Ting AY. "Live-cell mapping of organelle-associated RNAs via proximity biotinylation combined with protein-RNA crosslinking." ***eLife*** (2017).

Choi J, Lee S, **Mallard W**, Clement K, Tagliazucchi GM, Lim H, Choi IY, Ferrari F, Tsankov A, Pop R, Lee G, Rinn JL, Meissner A, Park P, Hochedlinger K. "A comparison of genetically matched cell lines reveals the equivalence of human iPSCs and ESCs." ***Nature Biotechnology*** (2015).

Cancer Genome Atlas Research Network. "Comprehensive molecular profiling of lung adenocarcinoma." ***Nature*** (2014).

Wishnow EH, **Mallard W**, Ravi V, Lockwood S, Fitelson W, Wertheimer D, Townes CH. "Mid-infrared interferometry with high spectral resolution." ***Proc SPIE*** (2010).

Pearson C, Leibrandt D, Bakr W, **Mallard W**, Brown K, Chuang I. "Experimental Investigation of Planar Ion Traps." ***Physical Review A*** (2006).

Full list available on Google Scholar.